

Accessibility Percolation with Rough Mount Fuji labels

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Overview

- 1 Rough Mount Fuji Model
- 2 RMF accessibility percolation on Bienaymé-Galton-Watson trees
- 3 RMF on the lattice

Accessibility percolation

- *Accessibility percolation* is a model inspired by evolutionary biology
- assigns random *fitness* values to each vertex in a graph.
- A path is considered *accessible* if the fitness values along it are strictly increasing,
- and *accessibility percolation* occurs when an accessible path exists between the root or origin to ∞ (or a target vertex for finite graphs).

Rough Mount Fuji model

Definition (Rough Mount Fuji)

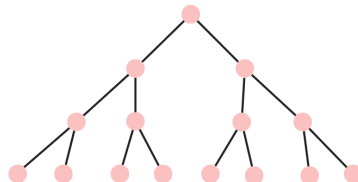
Fix $\theta \in [0, 1]$, and let $\{U_v : v \in V\}$ be a collection of independent $\text{Uniform}(0, 1)$ random variables indexed by the vertex set. For each $v \in V$, define

$$X_v = U_v + \theta d(\rho, v)$$

where $d(\rho, v)$ is the distance from the root to vertex v .

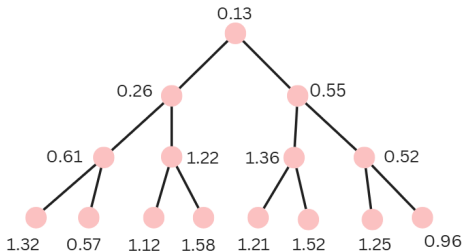
Rough Mount Fuji accessibility percolation on an m-ary tree

- Consider a regular tree:
- Every vertex is labelled with an i.i.d., $U[0, 1]$ plus a constant $\theta \in [0, 1]$ times the distance from the root to the vertex.



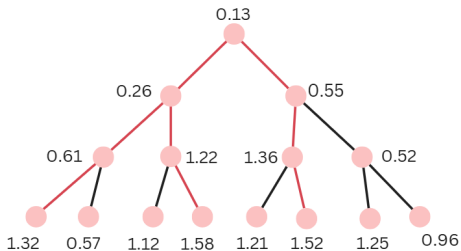
Rough Mount Fuji accessibility percolation on an m-ary tree

- Consider an m-ary tree.
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- We are looking for paths whose labels are in increasing order.



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Main Goal

- Bound the critical threshold θ_c
- such that there is no RMF accessibility percolation for $\theta < \theta_c$,
- whereas there is RMF accessibility percolation for $\theta > \theta_c$.

No RMF Accessibility Percolation on trees

Theorem (1)

Suppose that T is a Bienaymé-Galton-Watson tree whose offspring distribution L satisfies $\mathbb{E}[L] = m \in (1, \infty)$. For any $\theta \leq \frac{1}{em}$,

$$\mathbb{P}(\text{there exists an infinite increasing path}) = 0,$$

Upper bound on the probability of one path being increasing

For any $h \in \mathbb{N}$ and $\theta \in [0, 1]$,

$$\begin{aligned} \mathbb{P}(X_0 < X_1 < X_2 < \dots < X_h) \\ = \mathbb{P}(U_0 < U_1 + \theta < U_2 + 2\theta < \dots < U_h + h\theta) \leq \frac{(1 + \theta h)^{h+1}}{(h + 1)!} \end{aligned}$$

Proof of Theorem 1

Let P_h be the set of paths from the root to generation h : $\#P_h = m^h$.

Let N_h be the set of *increasing* paths in P_h .

By Markov's inequality and Stirling's approximation

$$\begin{aligned}\mathbb{P}(\#N_h \geq 1) &\leq \mathbb{E}[\#N_h] = m^h \mathbb{P}(\text{one fixed path of length } h \text{ is increasing}) \\ &\leq m^h \frac{(1 + \theta h)^{h+1}}{(h+1)!} \\ &\leq \frac{1}{\sqrt{2\pi(h+1)}} \left(\frac{em}{h+1} + em\theta \right)^{h+1}\end{aligned}$$

In particular, if $\theta \leq \frac{1}{em}$, we have $\mathbb{P}(N_h \geq 1) \xrightarrow{h \rightarrow \infty} 0$

RMF accessibility percolation on all Bienaymé-Galton-Watson trees for larger θ

Theorem (2)

Suppose that T is a Bienaymé-Galton-Watson tree whose offspring distribution L satisfies $\mathbb{E}[L] = m \in (1, \infty)$. Then

$$\theta_c(m) \leq 1 - \sqrt{1 - \frac{1}{m}} \sim \frac{1}{2m}.$$

Exact characterisation of the critical RMF accessibility percolation threshold for all Bienaymé-Galton-Watson trees

Theorem (3)

For $\theta \in (0, 1]$ and $x \in \mathbb{R}$, define

$$Q_{\theta}(x) = \sum_{j=0}^{\lfloor 1/\theta \rfloor + 1} \frac{(-x)^j}{j!} (1 - (j-1)\theta)^j.$$

Let $m_c(\theta)$ be the minimal root of the polynomial equation

$$Q_{\theta}(m) = 0. \tag{1}$$

Suppose that T is a Galton-Watson tree whose offspring distribution L satisfies $\mathbb{E}[L] = m$ and $\mathbb{E}[L \log_+ L] < \infty$. There is Rough Mount Fuji accessibility percolation on T if and only if $m > m_c(\theta)$.

Bounds on the probability of one path being increasing

Corollary

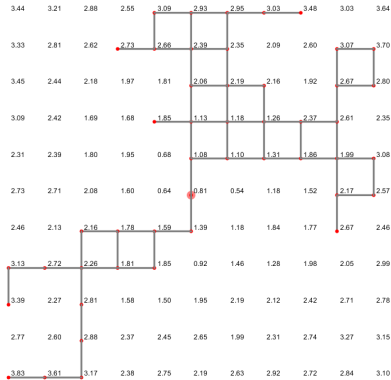
There exist constants $0 < c_1 < 1 < c_2 < \infty$ such that for any $\theta \in (0, 1)$,

$$\frac{c_1}{m_c(\theta)^h} \leq \mathbb{P}(U_0 < U_1 + \theta < \dots < U_h + h\theta) \leq \frac{c_2}{m_c(\theta)^h},$$

where $m_c(\theta)$ is the minimal root of the polynomial equation (1), and in particular

$$\frac{1}{e\theta} \leq m_c(\theta) \leq \frac{1}{2\theta - \theta^2}.$$

Rough Mount Fuji on the lattice



■ $\theta = 0.43$

Rough Mount Fuji on the lattice

Paths:

- (a) Non-Backtracking: Paths $x_0, x_1, \dots$ where $x_0 = 0$ and $d(0, x_{j+1}) > d(0, x_j)$ for each j . (Each step moves further from the origin.)
- (b) All (Simple) Paths: No restriction on the path.

Distances ($d(x, y)$):

- (i) Graph Distance: $d(x, y) = \|x - y\|_1$
- (ii) L_p Distance: $d(x, y) = \|x - y\|_p, p > 1$

Rough Mount Fuji accessibility percolation on the 2d-lattice (L_1 distance, non-backtracking paths)



Figure: left $\theta = 0.31$ and right
 $\theta = 0.33$

Rough Mount Fuji on the 2d- integer lattice (L_1 distance, non-backtracking paths)

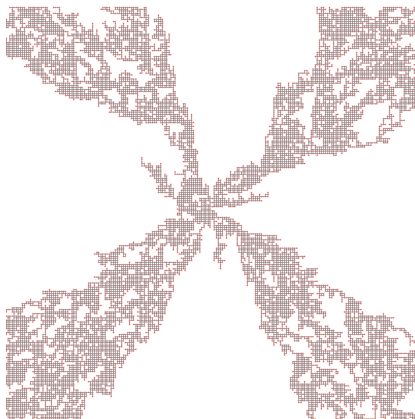


Figure: $\theta = 0.35$

Rough Mount Fuji on the 2d- integer lattice with non-backtracking paths

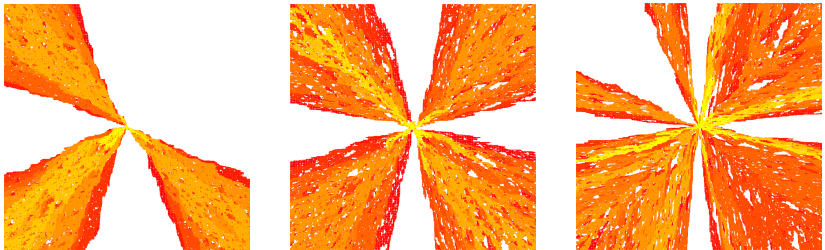


Figure: ℓ^1 distance, $\theta = 0.33$ (yellow) to 0.42 (red). ℓ^2 distance $\theta = 0.45$ (yellow) to 0.54 (red). ℓ^4 distance, $\theta = 0.53$ (yellow) to 0.62 (red).

Rough Mount Fuji on the 2d- integer lattice with backsteps

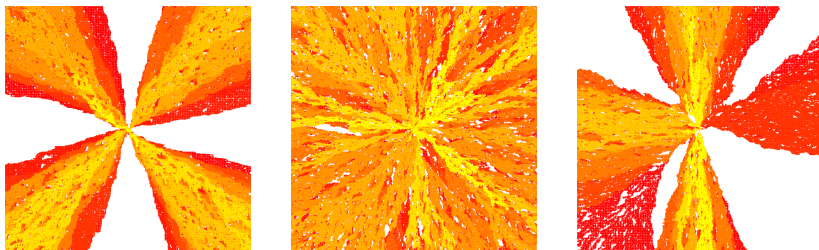


Figure: ℓ^1 distance, $\theta = 0.33$ (yellow) to 0.42 (red). ℓ^2 distance, $\theta = 0.45$ (yellow) to 0.54 (red). ℓ^4 distance, $\theta = 0.48$ (yellow) to 0.57 (red).

No RMF accessibility percolation on the d-lattice for small θ

Theorem (4)

For any $d \geq 2$, there is no Rough Mount Fuji accessibility percolation on the d -dimensional lattice with any L_p distance, $p \geq 1$:

- with non-backtracking paths when $\theta \leq \frac{1}{de}$;
- with all paths when $\theta \leq \frac{1}{(2d-1)e}$.

RMF accessibility percolation on the d -lattice for larger θ

Theorem (5)

There is Rough Mount Fuji accessibility percolation on the d -dimensional lattice, above θ_c for some $\theta_c < 1$

Thank you for your attention!

